Roland Singh

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Education

Indian Institute of Technology, Madras — **May 2024 – April 2028**
Bachelors of Science in Data Science and Programming

Sikkim Manipal Institute of Technology, Majitar — **July 2023 – June 2027**
Bachelors of Technology in Computer Science and Engineering

Don Bosco Senior Secondary School, Guwahati — **April 2021 – March 2023**
AISSCE (Class 12)

Don Bosco Senior Secondary School, Guwahati — **April 2009 – March 2021**
AISSE (Class 10)

Experience

Research Intern — **July 2026 - Present**
The University of Salford, Manchester, United Kingdom — *Remote*

- Collaborate with Prof. Shivakumara Palaiahnakote on scene text recognition in complex, unconstrained visual environments as part of an international research initiative.
- Develop sequence modeling architectures for high-fidelity text extraction from visually complex scenes, optimizing feature representations for robustness under real-world imaging conditions.

Research Intern — **July 2026 - Present**
Indian Statistical Institute, Kolkata — *Hybrid*

- Conduct research under Prof. Umapada Pal in the Computer Vision and Pattern Recognition Unit, developing deep learning methods for restoring and recognizing text in severely degraded, unstructured document images.
- Design and benchmark novel restoration architectures that bypass traditional segmentation-based pipelines, targeting improved feature extraction and recognition accuracy on low-quality document scans.

Summer Intern — **May 2026 - Present**
Indian Institute of Technology, Kharagpur — *Onsite*

- Selected for IIT Kharagpur's GRISHMA Research Fellowship Program, working on deep learning frameworks for imaging inverse problems.
- Implement TV and Fields-of-Experts regularizers, solving the lower-level reconstruction with Adam and computing upper-level gradients via the implicit function theorem.
- Design an input-convex neural network (ICNN) based regularizer, benchmarking it on MNIST classification, CT and MRI segmentation.

Research Assistant — **Aug 2025 - Present**
Sikkim Manipal Institute of Technology, Majitar — *Onsite*

- Conduct research under Dr. Ashis Pradhan and Dr. Palash Ghosal spanning image restoration, underwater computer vision, and 3D medical image segmentation, developing physics-informed and geometry-aware deep learning architectures for degraded and volumetric imaging data.
- Designed Turbid-GAN, a lightweight (9.1M parameter) residual encoder-decoder with Triplet Attention for real-time underwater image enhancement, achieving a 234-fold improvement in ORB feature matching to enable reliable visual odometry for AUVs in sediment-degraded environments.
- Developed a Delta-Mamba framework integrating Deep Delta Learning (DDL) modules into a U-Mamba backbone for 3D medical image segmentation, evaluated on the ACDC and Synapse datasets.

Summer Intern — May 2025 - July 2025
Indian Institute of Technology, Guwahati — *Hybrid*

- Worked under Prof. Natesan Srinivasan to address critical challenges in low-light computer vision, developing solutions applicable to surveillance systems.
- Developed deep learning models for facial emotion recognition and object detection in low-light conditions, achieving 81% validation accuracy for emotion classification and 0.75 maP for object detection on challenging nighttime datasets.
- Implemented and evaluated multiple image enhancement techniques including CLAHE, Diffusion Models, and Zero-DCE++ to optimize preprocessing pipelines for downstream computer vision tasks.
- Fine-tuned YOLO v8 and pretrained CNNs (VGG16, ResNet50, EfficientNet) on custom datasets, integrating enhancement frameworks to improve detection performance in poorly illuminated environments.

Summer Intern — June 2025
National University of Singapore, Singapore — *Onsite*

- Conducted research in computer vision for autonomous systems under Dr. Amirhassan Monajemi, implementing state-of-the-art detection and tracking algorithms for critical infrastructure monitoring.
- Developed an autonomous computer vision system for underwater pipeline inspection using YOLOv8 and DeepSORT, enabling real-time detection and tracking of corrosion and structural damage on subsea infrastructure.
- Processed and annotated 1,775 frames from underwater video footage, implementing HSV segmentation and data augmentation techniques to train robust detection models under challenging aquatic conditions.

Publications

- **ClearVision: A Physics-Informed Lightweight GAN for Real-Time Enhancement of Turbid Underwater Imagery** (Accepted for Oral Presentation at *IEEE OCEANS 2026*, Sanya)
  R. Singh, *D. Das, A. Pradhan*
- **DDL-UMamba: Geometric Residual Learning for 3D Medical Image Segmentation** (Under Peer Review at Knowledge-Based Systems (Elsevier)
  D. Das, ***R. Singh***, *P. Ghosal, A. Pradhan*
- **Benchmarking GAN Architectures for Underwater Imaging: The Trade-off Between Model Compactness and Real-Time Latency** (Accepted for Presentation at IEEE GCON, 2026)
  D. Das, ***R. Singh***, *A. Pradhan, P. Ghosal*

Professional Affiliations

Student Member — Jan 2026 – Present
Institute of Electrical and Electronics Engineers (IEEE), Kolkata Chapter

- Member, IEEE Ocean Engineering Society.

Reviewer
IEEE Guwahati Subsection Conference (GCON), 2026

- Served as a reviewer for the peer-review process.

Co-Curricular Activities

Member — Dec 2024 - Present
IAESTE LC Manipal, Karnataka — *Remote*

- Actively involved in the International Association for the Exchange of Students for Technical Experience (IAESTE) Local Committee in Manipal, helping facilitate international internships for students.
- Received an offer to intern in Tunisia as part of the IAESTE Exchange Programme.

Projects

Women's Safety App **Aug 2024 - Nov 2025**

- Developed a dual-component safety system combining a reactive Android app (SOS alerts, shake detection, emergency navigation) with a proactive ML engine that predicts area-wise safety scores using XGBoost and NLP-based crime data extraction.
- Built an NLP pipeline using Bidirectional LSTM to automatically extract and classify crime incidents from 5,000+ news articles, achieving 84.8% accuracy, and trained an XGBoost model with 99% accuracy to predict safety levels across geographic regions.
- Implemented "Safest Route" simulation algorithm that intelligently navigates users around high-risk zones using Google Maps API.

Skills

- **Programming Languages:** Python, C, C++, MATLAB
- **Libraries and Frameworks:** PyTorch, TensorFlow, Transformers, Keras, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
- **Tools and Systems:** Git, Linux Shell (Bash), CUDA, HPC GPU Clusters
- **Development Environment:** Jupyter Notebook, Visual Studio Code, PyCharm
- **Databases:** MySQL, MongoDB
- **Miscellaneous:** LaTeX, Blender

Relevant Courses

- Machine Learning
- Deep Learning
- Computer Vision
- Digital Image Processing
- Soft Computing
- Data Structures in C++ and Python

Honors and Awards

- Awarded a 40% tuition fee waiver scholarship at Sikkim Manipal Institute of Technology based on JEE rank (2023).
- Black Belt in Taekwondo (ITF) and certified national instructor, with active participation in national and international workshops, tournaments, and events.